

Comparison of speed and performance between Python and R implementations of the AddiVortes algorithm

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1 Introduction

The purpose of this article is to compare the speed and performance of the AddiVortes algorithm between the current up-to-date Python and R implementations. We will benchmark the fitting time of a model, as well as the amount of time required to make simple predictions on a test set, to try to determine which of the two implementations is faster to use. We will also benchmark the in-sample and out-of-sample RMSE on that data to determine which of the two implementations gives predictions with lower errors.

The code used to generate the benchmarking data, as well as the generated data itself and all referenced files, are available to view at <https://github.com/snowy-the-leopard/AddiVortes>. This data was generated using the ADDIVORTES R package, version 0.6.6, available to view here <https://github.com/johnpaulgosling/AddiVortes>, and the ADDIVORTES Python package, version 0.6.6, available to view here <https://github.com/johnpaulgosling/py-addivortes>.

2 Benchmarking methodology

To ensure fairness, consistency, and accuracy with these tests, the following controls were implemented on these benchmarks:

1. The benchmarks were run on a new GitHub Codespace, a clean installation of Python and R were used, with only the necessary packages downloaded to minimise background resource usage. The Codespace used is a 2-core 2.8GHz system with 8GB of memory. Exact details of hardware used can be found at [AddiVortes\mySpecs.html](#).
2. The code was set up such that each implementation used only a single thread, to counteract the potentially different methods of multi threading between the two languages. This choice was made as we are working under the assumption that when we are working with HPC resources, the code will be optimised to allow multiple threads to be used at once. This choice allows us to compare algorithmic efficiency in each implementation more directly.
3. The same models and predictions were fitted repeatedly in order to capture the small variations in fit and prediction times between runs. This will also produce a set of slightly varying prediction results due to the random nature of the MCMC fitting algorithm, and allow us to better analyse how consistent our results are.
4. The benchmarks were run sequentially rather than simultaneously to minimise the risk of shared resources skewing the results, where one script could take computational priority over another, leading to a larger discrepancy in results.
5. The timings were taken only between the computation required for fitting and generating predictions, with the pre-generated datasets already loaded in prior to this.

3 Datasets used in benchmarking

To allow for a wide scope of comparison between the two implementations, the following 4 different datasets were used in benchmarking:

3.1 Boston dataset

The Boston Housing Dataset, derived from information from the U.S. Census Service. This consists of 13 numerical covariates, and we model it using the following fixed parameters, with the rest of the parameters taking their default values: `m = 50`

3.2 Synthetic dataset

We have constructed a synthetic dataset using noisy functions of independently and identically distributed $U[0, 1]$ observations. This consists of 5 numerical covariates, and we model it by taking all the parameters as their default values.

3.3 Spherical dataset

We have constructed random longitudes and latitudes to represent positions on a sphere, and a noisy response function. This consists of 2 numerical covariates (the coordinates), and we model it using the following fixed parameters, with the rest of the parameters taking their default values: `m = 50`, `totalMCMCiter = 500`, `mcmcBurnIn = 100`, `metric = "S"`. Note that this uses great circle distance, the shortest distance between two points over the surface of the sphere.

3.4 Categorical dataset

We have constructed this dataset using random numerical and categorical covariates, as well as a noisy response function. This consists of 2 numerical and 2 categorical covariates, and we model it using the following fixed parameters, with the rest of the parameters taking their default values: `m = 50`, `totalMCMCiter = 500`, `mcmcBurnIn = 100`, `catScaling = 1`

4 Benchmark results

The full benchmarks, which carry out model fitting and prediction on the above 4 datasets, were run for 500 iterations each. The code can be found at `AddiVortes\tests.R` and `AddiVortes\tests.py`

Table 1: Mean Fit and Prediction Times (R vs Python)

Test	Fit Mean (R)	Fit Mean (Python)	Fit % Diff	Pred Mean (R)	Pred Mean (Python)	Pred % Diff
Test 1	11.517	11.177	-2.954	6.984	3.844	-44.961
Test 2	1.404	1.253	-10.762	1.167	0.656	-43.764
Test 3	3.067	3.045	-0.713	2.219	1.973	-11.084
Test 4	0.508	0.402	-20.901	0.426	0.208	-51.005

Table 2: Standard Deviation of Fit and Prediction Times (R vs Python)

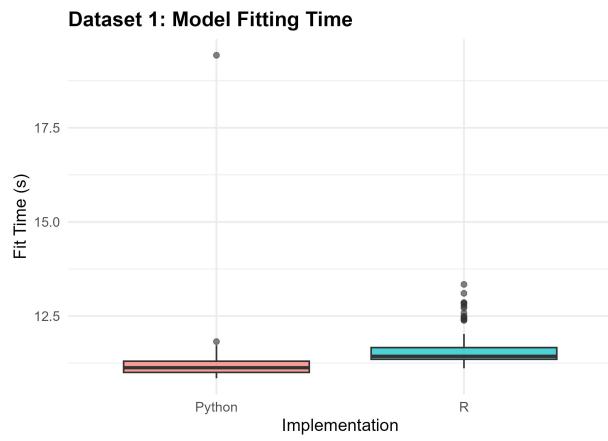
Test	Fit SD (R)	Fit SD (Python)	Fit % Diff	Pred SD (R)	Pred SD (Python)	Pred % Diff
Test 1	0.276	0.410	48.353	0.238	0.090	-62.180
Test 2	0.052	0.129	147.862	0.059	0.037	-37.148
Test 3	0.132	0.270	103.698	0.214	0.182	-15.142
Test 4	0.087	0.075	-14.421	0.039	0.030	-23.925

Table 3: Mean In-sample and Out-of-sample RMSE (R vs Python)

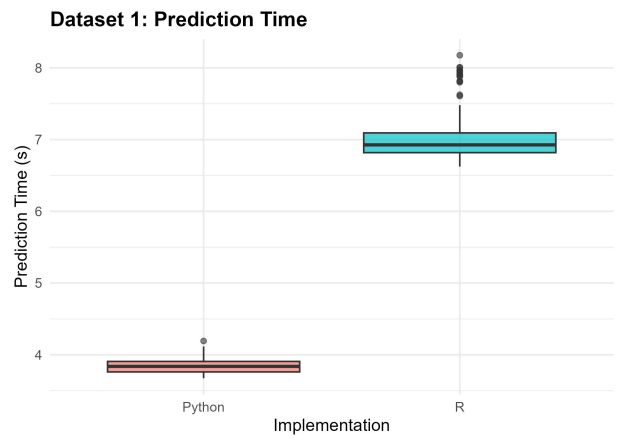
Test	iRMSE Mean (R)	iRMSE Mean (Python)	iRMSE % Diff	oRMSE Mean (R)	oRMSE Mean (Python)	oRMSE % Diff
Test 1	0.779	0.794	1.958	3.674	3.677	0.076
Test 2	1.020	1.081	6.003	1.360	1.420	4.401
Test 3	1.219	1.240	1.693	2.203	2.217	0.637
Test 4	2.237	2.244	0.341	3.036	3.037	0.029

Table 4: Standard Deviation of RMSE (R vs Python)

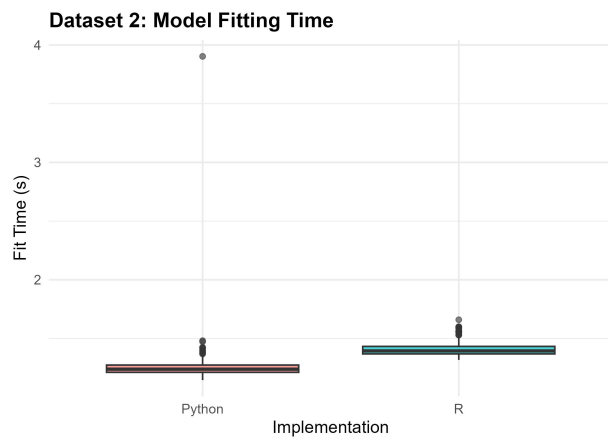
Test	iRMSE SD (R)	iRMSE SD (Python)	iRMSE % Diff	oRMSE SD (R)	oRMSE SD (Python)	oRMSE % Diff
Test 1	0.034	0.036	6.616	0.104	0.115	10.987
Test 2	0.082	0.082	-0.332	0.116	0.107	-8.077
Test 3	0.055	0.059	8.739	0.048	0.049	3.290
Test 4	0.037	0.037	0.417	0.067	0.071	6.515



(a) Fit 1



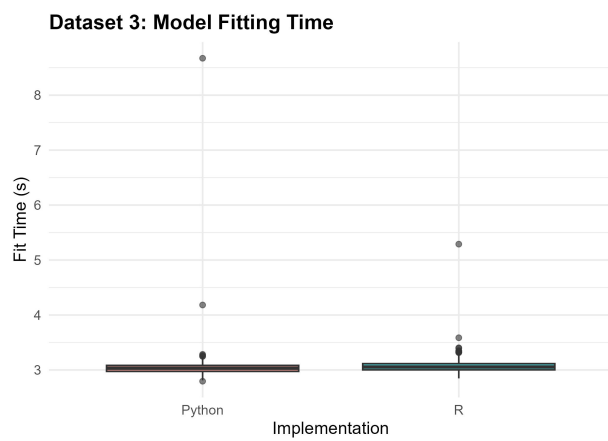
(b) Pred 1



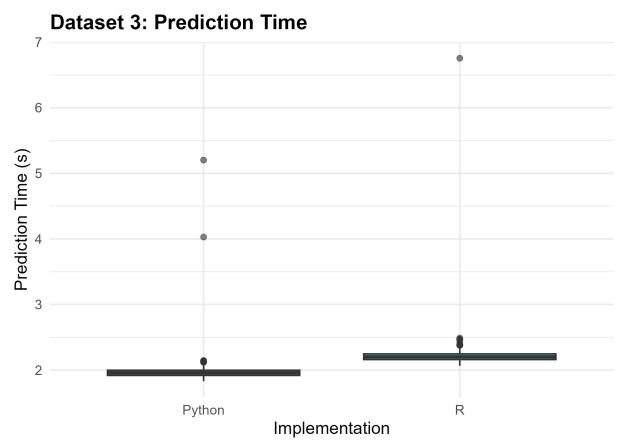
(c) Fit 2



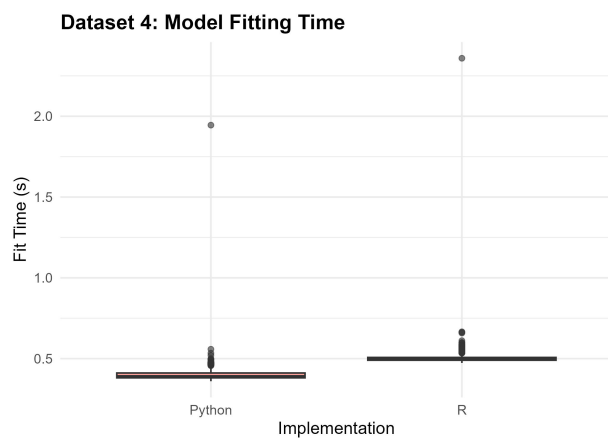
(d) Pred 2



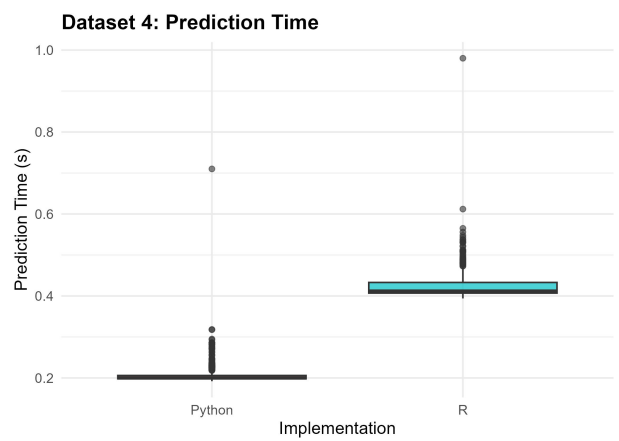
(e) Fit 3



(f) Pred 3

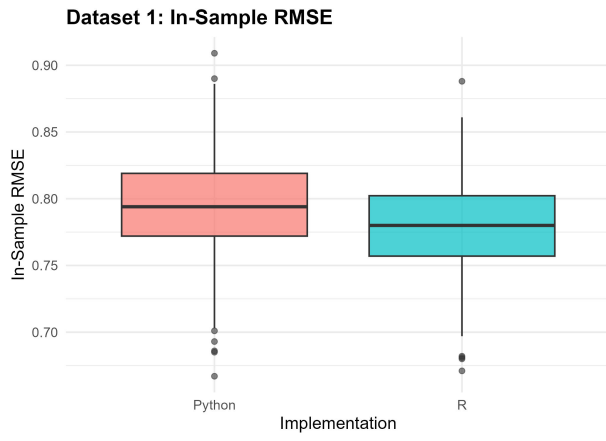


(g) Fit 4

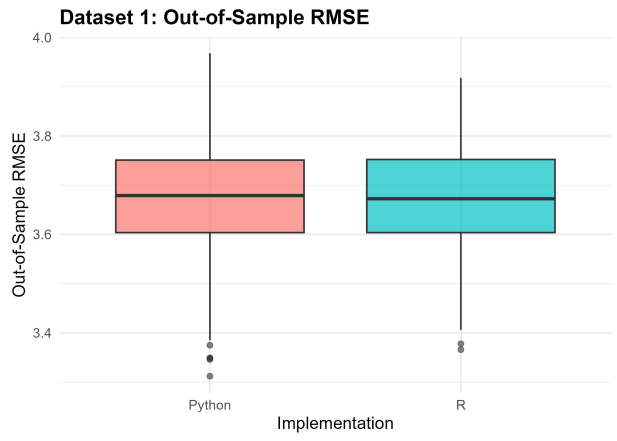


(h) Pred 4

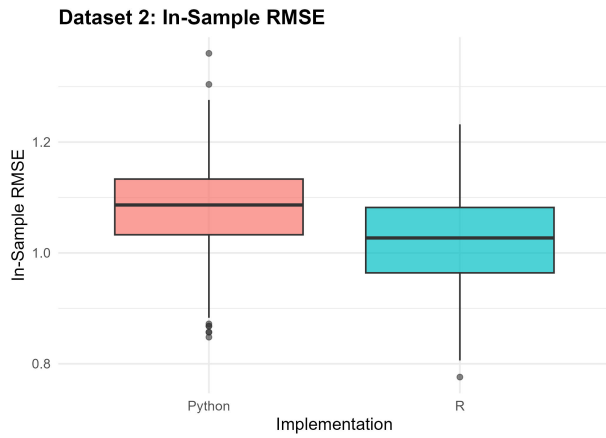
Figure 1: Model fitting and prediction times across datasets.



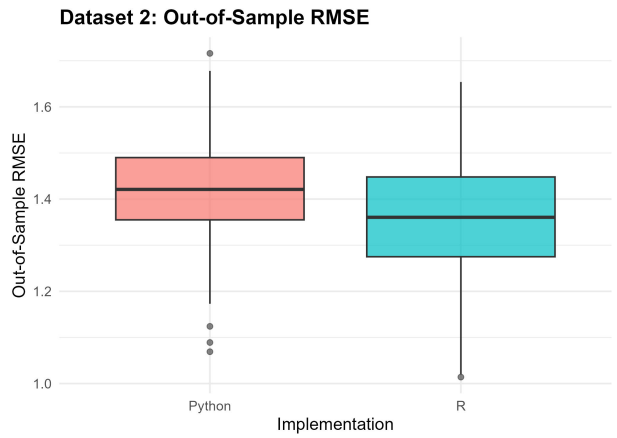
(a) iRMSE 1



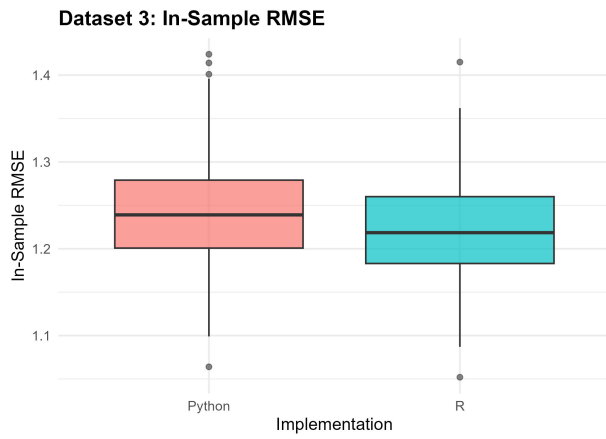
(b) oRMSE 1



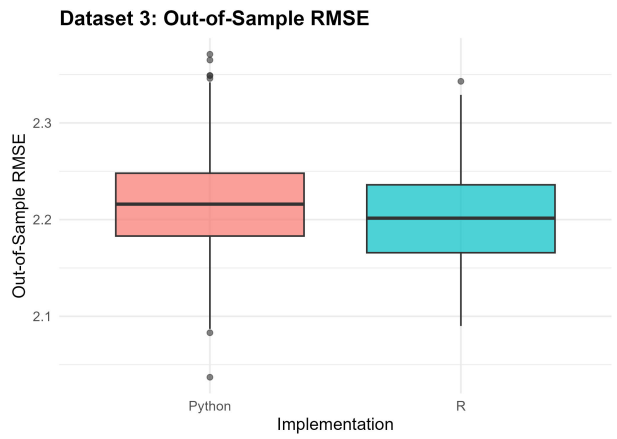
(c) iRMSE 2



(d) oRMSE 2



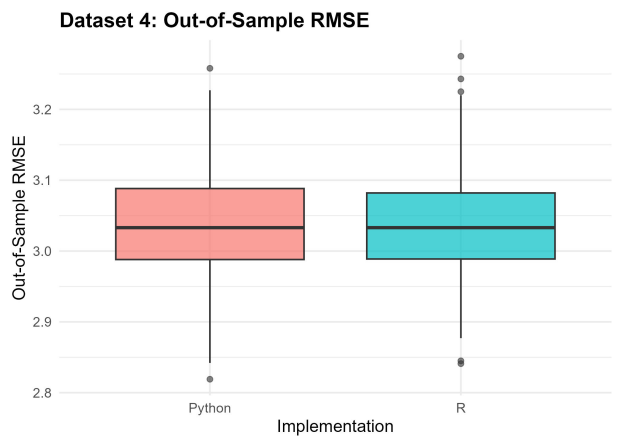
(e) iRMSE 3



(f) oRMSE 3



(g) iRMSE 4



(h) oRMSE 4

Figure 2: In-sample and out-of-sample RMSE across datasets.

5 Preliminary analysis of results

5.1 Fit times

The mean fit time using Python was substantially lower for all models. With an average decrease of 9% in mean fit time using Python, we can certainly conclude that the Python implementation is significantly faster than R at fitting the AddiVortes model using a Euclidean metric. The standard deviation of fit times using Python gave mixed results when compared to using R, but for the first 3 models Python has much higher standard deviations for fit time.

Hence, Python produces lower but less consistent fit times than R.

5.2 Prediction times

The mean prediction time using Python was significantly lower for all models, with an average decrease of 38% when using Python instead of R. The standard deviation of prediction times using Python was significantly lower for all models, with an average decrease of 35% when using Python instead of R.

Therefore in terms of both lower and more consistent prediction times, Python is superior.

5.3 In-sample RMSEs

The mean in-sample RMSE using Python was slightly higher for all models, with an average increase of 2% when using Python instead of R. The standard deviation of in-sample RMSEs using Python were on average 4% higher than when using R.

Therefore it is unlikely there is a significant difference between in-sample RMSEs between R and Python.

5.4 Out-of-sample RMSEs

The mean out-of-sample RMSE using Python consistently gave slightly higher values, with an average increase of 1% across the 4 models. The standard deviation of out-of-sample RMSEs using Python gave mixed results, but overall seemed to be higher, with an average increase of 3% across the 4 models.

Therefore it is unlikely there is a significant difference between out-of-sample RMSEs between R and Python

6 Significance of results using standard errors

Let $\bar{X}_{f_1}$ be our sample mean fit time for test 1 using R and $\bar{Y}_{f_1}$ be our sample mean fit time for test 1 using Python.

$$Var(\bar{X}_{f_1} - \bar{Y}_{f_1}) = Var(\bar{X}_{f_1}) + Var(\bar{Y}_{f_1})$$

Note that we are using samples with 500 observations and hence need to use our standard errors, by scaling our standard deviations from tables 2 and 4 by a factor of $\frac{1}{\sqrt{n}}$. We can then calculate the standard error of the difference of the means:

$$SE(\bar{X}_{f_1} - \bar{Y}_{f_1}) = \sqrt{SE(\bar{X}_{f_1})^2 + SE(\bar{Y}_{f_1})^2}$$

We divide the difference in the mean values $\bar{X}_{f_1} - \bar{Y}_{f_1}$ by the combined standard error to give z-values for these differences. We will consider z-values over 2 as significant (roughly 95% significance level). The results of these analyses are below:

Table 5: Fit and prediction standard errors

Test	Fit SE (R)	Fit SE (Python)	Pred SE (R)	Pred SE (Python)	Fit SE (Comb)	Pred SE (Comb)
Test 1	0.012	0.018	0.011	0.004	0.022	0.011
Test 2	0.002	0.006	0.003	0.002	0.006	0.003
Test 3	0.006	0.012	0.010	0.008	0.013	0.013
Test 4	0.004	0.003	0.002	0.001	0.005	0.002

Table 6: iRMSE and oRMSE standard errors

Test	iRMSE SE (R)	iRMSE SE (Python)	oRMSE SE (R)	oRMSE SE (Python)	iRMSE SE (Comb)	oRMSE SE (Comb)
Test 1	0.002	0.002	0.005	0.005	0.002	0.007
Test 2	0.004	0.004	0.005	0.005	0.005	0.007
Test 3	0.002	0.003	0.002	0.002	0.004	0.003
Test 4	0.002	0.002	0.003	0.003	0.002	0.004

Table 7: Fit and prediction z-values

Test	Fit Diff	Pred Diff	Fit SE (Comb)	Pred SE (Comb)	Fit Diff SEs	Pred Diff SEs
Test 1	0.340	3.140	0.022	0.011	15.405	275.972
Test 2	0.151	0.511	0.006	0.003	24.256	164.673
Test 3	0.022	0.246	0.013	0.013	1.626	19.556
Test 4	0.106	0.217	0.005	0.002	20.725	98.798

Table 8: iRMSE and oRMSE z-values

Test	iRMSE Diff	oRMSE Diff	iRMSE SE (Comb)	oRMSE SE (Comb)	iRMSE Diff SEs	oRMSE Diff SEs
Test 1	0.015	0.003	0.002	0.007	6.933	0.402
Test 2	0.061	0.060	0.005	0.007	11.827	8.468
Test 3	0.021	0.014	0.004	0.003	5.713	4.576
Test 4	0.008	0.001	0.002	0.004	3.256	0.204

Clearly, the vast majority of the differences in fit times and prediction times are highly significant, with Python having the significantly lower times for both. Some of the iRMSE and oRMSE differences are significant, but there are also some insignificant results, especially within oRMSE where significances were lower than in iRMSE.

Therefore, since there is limited evidence to suggest the R implementation has lower oRMSEs, and there is strong evidence to suggest the Python implementation has faster fit and prediction times, I would suggest Python to be used for further implementation and investigation